

Quantum Mechanics I

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Welcome to Quantum Mechanics I. This course is the first of a two-semester sequence on quantum mechanics. Please see the syllabus for information regarding the course structure, grading, and policies.



Reading: Please note that while I will be providing some typed notes and exercises every now and then, this will be the exception rather than the rule. Instead, expect your main reference to be the textbook for this course (Sakurai and Napolitano). Any assigned reading announcements will be made on ICON.

One excellent complement to Sakurai is **Principles of Quantum Mechanics** by R. Shankar. Chapters 1 through 6 are particularly useful as a review of the basics of quantum mechanics. If you feel utterly confused about quantum mechanics, I strongly recommend to take a few evenings or a weekend to read all first six chapters of Shankar's book.

Griffiths' "Introduction to Quantum Mechanics" is also a good but more basics reference. It is very useful, however, for some topics like the Hydrogen atom.

If you feel bored, I recommend Weinberg's **Lectures on Quantum Mechanics** and Feynman's **Lectures on Physics**, Volume III. Please do not hesitate to ask me for more suggestions.

For a full reading list and suggested textbooks, please see the **Reading List on ICON**. I will update the reading list as we progress through the course as necessary.



Exercises: There will be two types of exercises: homework assignments and in-class exercises. Homework assignments are graded and count towards 30% of your final grade. In-class exercises are *not* graded and are meant to help you understand what we are covering in class and initiate discussions.

You are encouraged to work together on both in-class and homework exercises.

Understanding in-class exercises and homeworks will make you more prepared for the exams.

I do not expect you to derive difficult integrals from scratch (many will show up in the Hydrogen atom), please use **Mathematica** or other computational tools.



Calendar: Below you will find some important dates and a tentative calendar of topics we will cover throughout the semester. This is also available in the course **syllabus** (available on ICON).

Important Dates

- 1st Midterm: Wednesday, Oct 14th, 2026
 - 2nd Midterm: Wednesday, Nov 18th, 2026
 - Final exam: **TBD**, December **TBD**, 2026
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Tentative Topics Calendar

Please note this calendar is a tentative plan and subject to change. Ranges are inclusive.

- **Module 1: Introduction and Formalism**
 - **Topics:** Dirac notation and operators, measurements, position and momentum, and wavefunctions.
 - **Dates:** Weeks 1 - 3 (Aug 24 - Sep 11)
 - **Note:** Labor Day (No class Sep 7)
- **Module 2: Time Evolution**
 - **Topics:** Time-evolution operator, Schrödinger's equation and solutions, harmonic oscillator, propagators, and path integrals.
 - **Dates:** Weeks 4 - 8 (Sep 14 - Oct 16)
- **Midterm 1:** Wednesday, October 14
- **Module 3: Angular Momentum**
 - **Topics:** Rotations, spin and angular momentum operators, hydrogen atom, addition of angular momentum, tensors, and the Wigner - Eckart theorem.
 - **Dates:** Weeks 9 - 11 (Oct 19 - Nov 6)
- **Module 4: Ensembles and Density Matrix**
 - **Topics:** Density operators, product and mixed states, and open systems.
 - **Dates:** Weeks 12 - 13 (Nov 9 - Nov 20)
 - **Note:** Fall Break during Week 14 (No classes Nov 22 - 29)
- **Midterm 2:** Wednesday, November 18
- **Module 5: Symmetries & Overflow**
 - **Topics:** Symmetries in central potentials, parity and selection rules, and time reversal.
 - **Dates:** Weeks 15 - 16 (Nov 30 - Dec 11)
- **Finals Week:** Dec 14 - 18

Review Resources

I taught QM-II last semester, and I will reproduce here what I think students needed to know in order to succeed in QM-II. This way you know what to make sure you should know by the end of this semester to be well-prepared for the next.

In general, you should be familiar with:

- **Dirac notation** and the concept of **Hilbert space**,
- **time-evolution** in Dirac and Schrödinger pictures,
- the concepts of **generators** and **conserved quantities**,
- theory of **angular momentum**, selection rules, and the addition of angular momenta,
- the **1D scattering problem** with transmission and reflection coefficients,
- the **Hydrogen atom** and conserved quantities in central potentials,
- the **harmonic oscillator** and raising and lowering operators, and
- all things related to **two-state systems**.



My previously recommended reading for students taking QM-II was:

- **Dirac notation and wavefunctions:**
 - Sakurai and Napolitano, Sections 1.2, 1.6, and 1.7. These must become second nature to you.
 - Additional reading: Shankar, Chapter 4 and 6 (you can read these entirely).
- **Time-evolution:**
 - Sakurai and Napolitano, Sections 2.1 and 2.4.
 - Schrödinger and Heisenberg pictures: Sakurai and Napolitano, Section 2.2.
 - See also the propagator approach in Sections 2.6 and the density matrix approach in Section 3.4.
 - Additional reading: Shankar, Section 4.3.
- **Standard Problems:**
 - Sakurai and Napolitano, parts of Section 2.5 but mostly the elementary problems in this chapter (free particle case, finite and infinite potential wells, and the harmonic oscillator).
 - Additional reading: Shankar, Chapter 5. If you understand “Section 5.4 - A Problem in Scattering” you are in good shape (we will do this in class and later revisit it with more machinery).